DAY 06 / 11 · WEB DEV IN 11 DAYS · GUIDED

Lists of shows

Arrays, objects, loops

ShowShelf after today: **small functions that pick shows from a list of twelve**

TODAY

The plan for the next 90 minutes

Minutes	Topic
10	Recap: five questions about Day 5
10	Arrays
10	Objects
8	The twelve shows in data.js
12	Loops, and picking with a loop
12	filter and map
8	find, and searching text
5	The big idea, and questions
15	This afternoon in the Lab

RECAP · DAY 5

Five quick questions

1. function sayHello() { … } or sayHello(); : which one defines?
2. greet(name) and greet("Ana"): parameter or argument?
3. Why did the cards say "undefined hours" at first?
4. rating >= 8 tested before rating >= 9: what goes wrong?
5. function double(n) { return n * 2; } as an arrow function?

ARRAYS

Many values, in order

Square brackets around the list, commas between the items.

The **index** is an item's position. Counting starts at **0**.

A const list can still grow. It just can't be replaced.

```
const networks = ["HBO", "NBC", "BBC Three"];

networks[0];       // "HBO"
networks[1];       // "NBC"
networks.length;   // 3

networks.push("AMC");
networks.length;   // 4
```

OBJECTS

One thing, with named values

Each **property** is a key, a colon, and a value.

A dot reads it. A dot and **=** change it.

> Inside an object: a colon, not =.

```
const tonight = {
  show: "Dark",
  episode: 3,
  watched: false,
};

tonight.show;            // "Dark"
tonight.watched = true;
```

THE TWELVE SHOWS

An array of objects: read from the outside in

`shows`
the list

→

`[6]`
the 7th item

→

`.episodes`
its property

```
shows[3].name;                    // "Game of Thrones"
shows[6].episodes;                // 5 (Chernobyl)
shows[shows.length - 1].network;  // "Nickelodeon"
```

LOOPS

Run the block once for every item

show holds the next item on each round.

Name it after the list: **show** of **shows**.

```
for (const show of shows) {
  console.log(show.network);
}
```

Three lines of code, twelve lines in the Console.

PICKING WITH A LOOP

Empty list, loop, if, push

After the loop, **hbo** holds only the shows that passed the test.

> The empty list goes before the loop, not inside it.

```
const hbo = [];
for (const show of shows) {
  if (show.network === "HBO") {
    hbo.push(show);
  }
}
console.log(hbo.length);   // 4
```

FILTER AND MAP

filter keeps some, map changes each

[12 shows]	filter(HBO) →	4 shows: **fewer items**
[12 shows]	map(network) →	12 networks: **same count**

```
function onHBO(list) {
  return list.filter((show) => show.network === "HBO");
}

shows.map((show) => show.network);   // ["NBC", "AMC", "NBC", "HBO", ...]
```

FIND

One item, not a list

find gives back the first item that passes the test.

Nothing matches? find gives undefined.

```
const pick = shows.find(
  (show) => show.id === 10
);

pick.name;   // "Band of Brothers"
```

Apps look things up by **id**: unique, and it never changes.

SEARCHING TEXT

trim, toLowerCase, includes

```
const typed = "  THRONES ";
const needle = typed.trim().toLowerCase();                  // "thrones"

"Game of Thrones".toLowerCase().includes(needle);          // true
shows[1].genres.includes("Crime");                         // true
```

trim: no spaces at the ends

toLowerCase: on both sides

includes: is it inside?

THE BIG IDEA

filter keeps some. map changes each. find gives one.

THIS AFTERNOON IN THE LAB

Ten steps, about three hours

Steps 1–3

The data

1 A list: arrays

2 One show: objects

3 The twelve shows

Steps 4–5

Loops

4 Go through every show

5 Keep only the best

Steps 6–8

Array methods

6 filter: the short way

7 map: one thing from each

8 find: one show by its id

Steps 9–10

Search, genres

9 Search by name

10 Browse by genre

Milestone: all five sections on the page are filled by your functions.

STUCK?

In this order

1

Read the message

The “Not yet” box says what to fix.

2

Console

A red line names the problem and the line number.

3

Hint

Two per step. The second is nearly the answer.

4

Answer

Compare it with yours. Find the difference.

5

Ask

Your teacher, or the person next to you.

Most of the time, steps 1 to 3 are enough. That's how developers work too.

END OF THE DAY

Before you go

1. favourites[1]: which item is it? And the last one?
2. filter, map or find: all names? only comedies? id 7?
3. Why is show.genres === "Comedy" never true?

FINISHED EARLY?

Extra challenges

Best first · Count the genres · Watch everything

TOMORROW

Make the page react

Build the cards yourself, and react to clicks and typing.